Λύσεις Ενδιάμεσης Εξέτασης 2019

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Άσκηση 1) $x = bt^2 + ct^3 = 1{,}4t^2 + 0{,}1t^3$ $\quad v_\mu = \frac{x(10)-x(0)}{10-0} = \frac{140+100}{10} = 24\ m/s$

$u = \frac{dx}{dt} = 2{,}8t + 0{,}3t^2 \Rightarrow u(0) = 0\ m/s$

$u(5) = 21{,}5\ m/s$

$u(10) = 58\ m/s$

Άσκηση 2) $m_A = 4kg$, $m_B = 25kg$, $\mu_K = 0{,}2$

i) Αν $a = 2\ m/s^2$ πόση είναι η μάζα του C.

A: $T_1 - m_A g = m_A a$

B: $N = m_B g$, $T_2 - T_1 - \mu_K m_B g = m_B a$

C: $m_C g - T_2 = m_C a$

Αθροίζοντας τις πιο πάνω τρεις εξισώσεις: $m_C - \mu_K m_B - m_A = (m_A + m_B + m_C)\, a/g$

$$\Rightarrow m_C = \frac{(m_A + \mu_K m_B)g + (m_A + m_B)a}{g - a} = 18{,}5\ kg$$

ii) $T_1 = m_A(a+g) = 48N$

$T_2 = m_C(g-a) = 148N$

iii) Από ερώτημα (i): $a = \frac{m_C - \mu_K m_B - m_A}{m_A + m_B + m_C} g \Rightarrow u = \sqrt{2gd\,\frac{m_C - \mu_K m_B - m_A}{m_A + m_B + m_C}} = 1\ m/s$

αν $m_C = 11\ kg$ (*)

Άσκηση 3) $\theta = 40{,}9°$, $h = 900m$, $t_{tot} = 65'$

i) άξονας x: $a_x = 0$ $\quad u_x = u_0 \cos\theta$ $\quad x = u_0 \cos\theta\, t$

άξονα y: $a_y = -g$ $\quad u_y = -u_0 \sin\theta - gt$ $\quad y = -u_0 \sin\theta\, t - \frac{1}{2} g t^2$

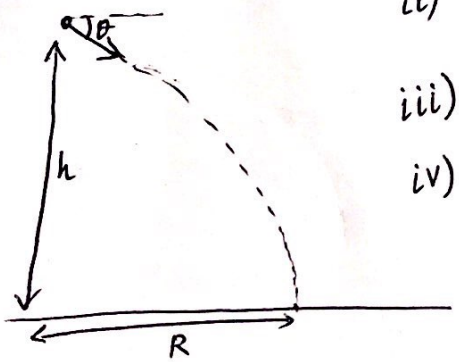
ii) $y = h$ $\quad t = t_{tot}$ $\quad -h = -u_0 \sin\theta\, t_{tot} - \frac{1}{2} g t_{tot}^2$

$$\Rightarrow u_0 = \frac{h - \frac{1}{2} g t_{tot}^2}{\sin\theta \cdot t_{tot}} = 183{,}29\ m/s$$

iii) $R = u_0 \cos\theta\, t_{tot} = 831{,}18\ m$

iv) $u_x = u_0 \cos\theta = 138{,}53\ m/s \Rightarrow \vec{u} = (138{,}53\,\hat{i} - 150{,}00\,\hat{j})\ m/s$

$u_y = -u_0 \sin\theta - g t_{tot} = -150\ m/s$



(*) Δεύτερος τρόπος: $K_1 + U_1 + W_{other} = K_2 + U_2$

$-m_A g l_A - m_C g l_C - \mu_K m_B g d = \frac{1}{2}(m_A + m_A + m_C)u^2 - m_A g(l_A - d) - m_C g(l_C + d)$

$$u = \sqrt{2gd\,\frac{m_C - \mu_K m_B - m_A}{m_A + m_B + m_C}} = 1\ m/s$$

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